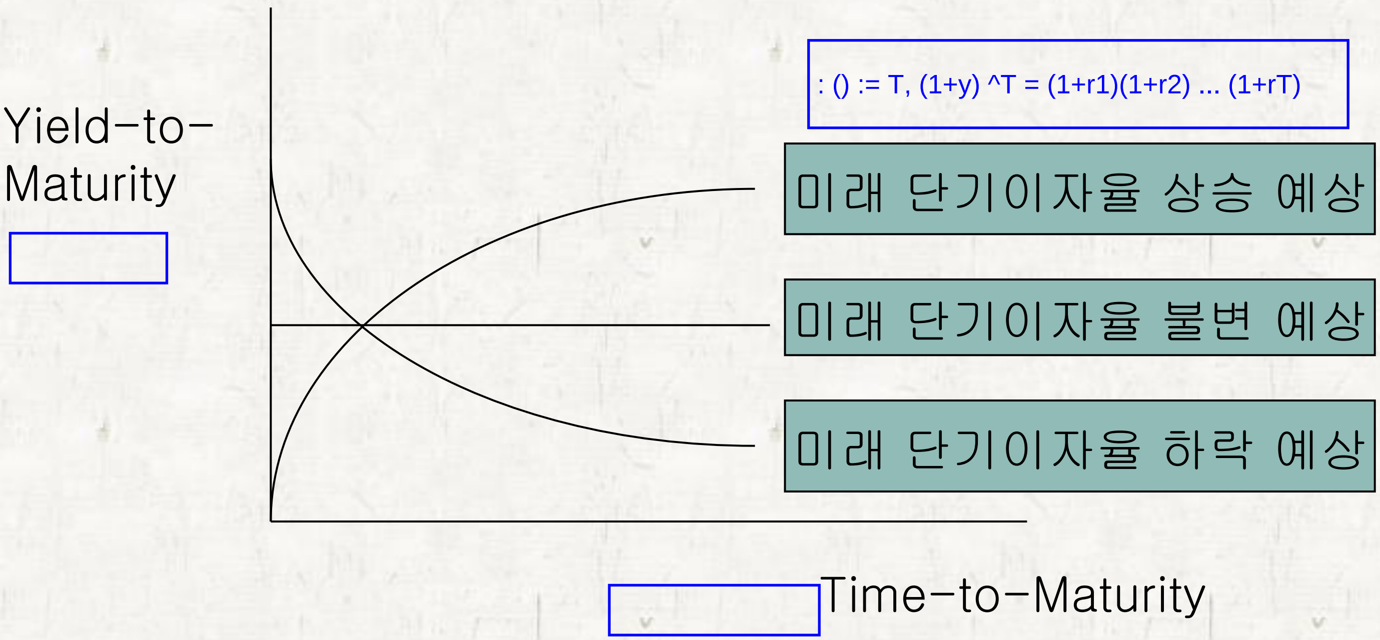
금융투자론

2025-1

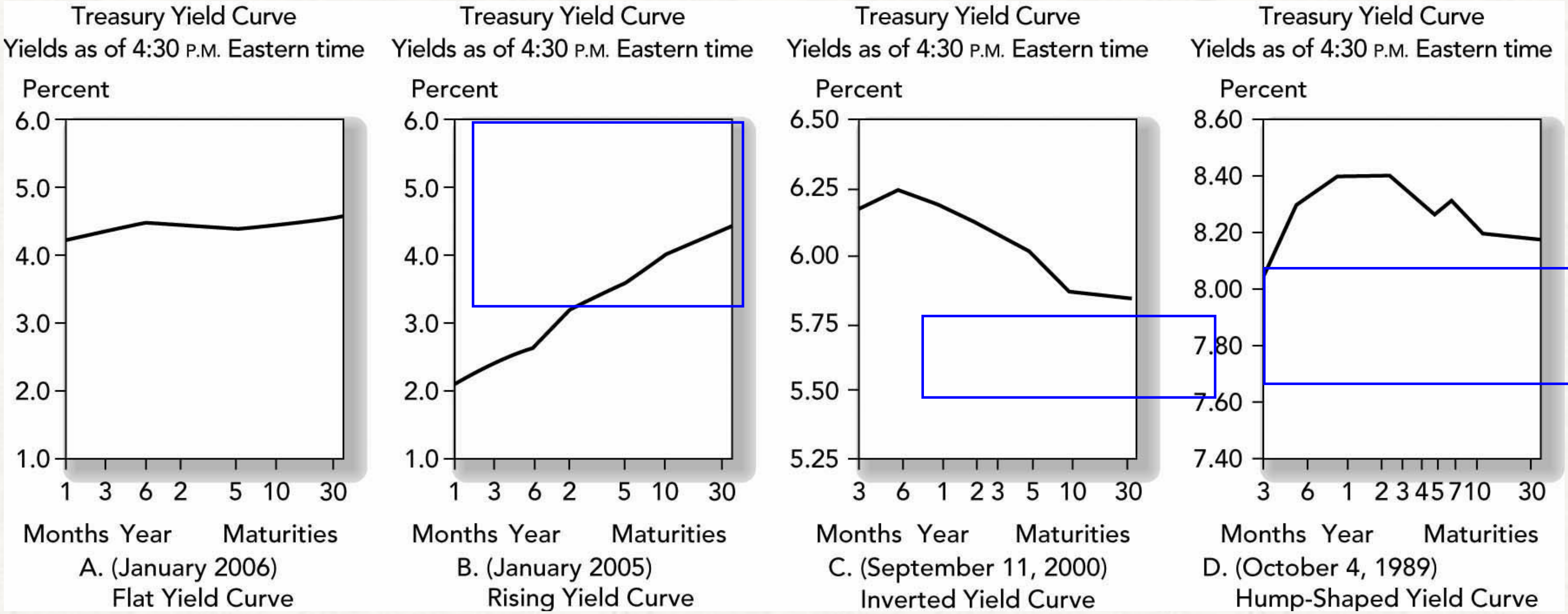
이자율 기간구조(Term structure of interest rates)

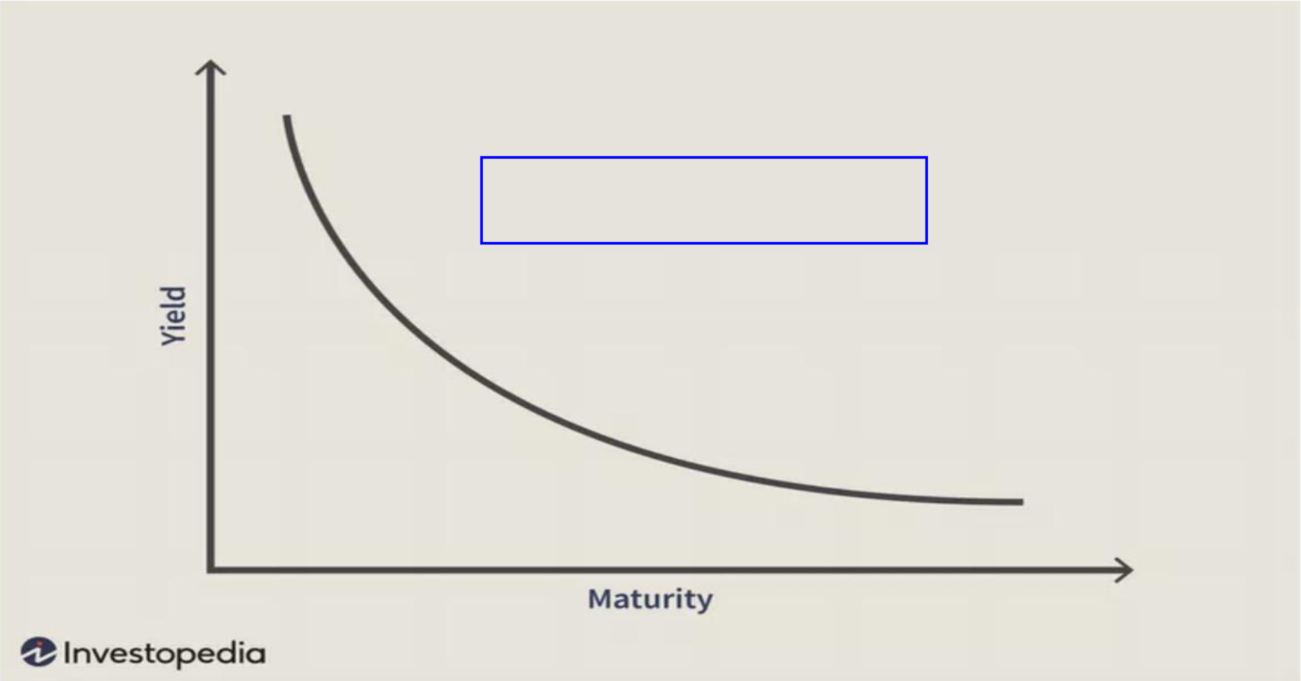
수익률곡선과 미래 이자율

- 수익률곡선(이자율 기간구조의 형태)는 미래 단기이자율에 대한 시장의 예상을 반영



Treasury Yield Curves





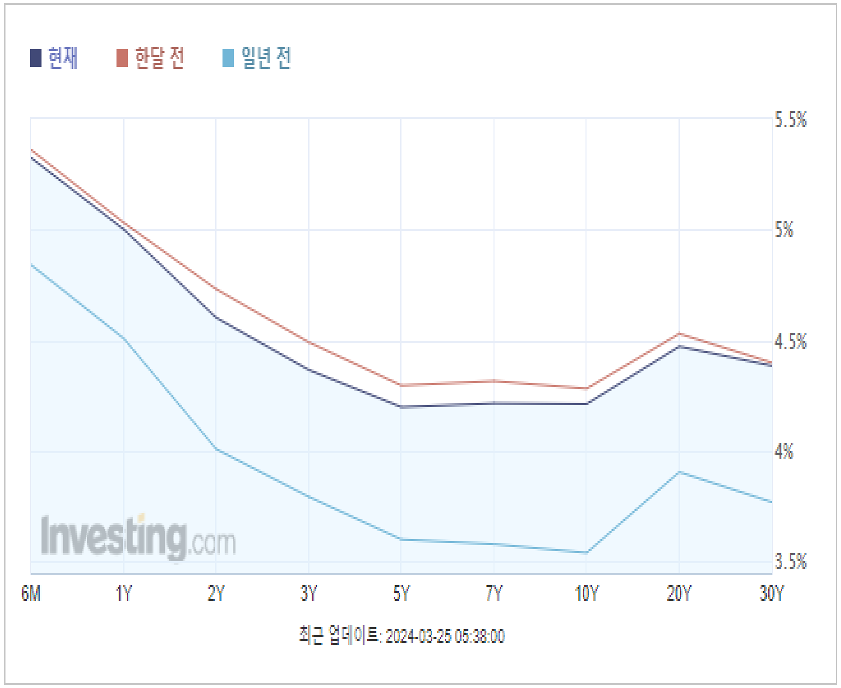
Inverted Yield Curve.
Image by Julie Bang © Investopedia 2019

A yield curve inverts when long-term interest rates drop below short-term rates, indicating that investors are moving money away from short-term bonds and into long-term ones. This suggests that the market as a whole is becoming more pessimistic about the economic prospects for the near future.

국채 수익률곡선

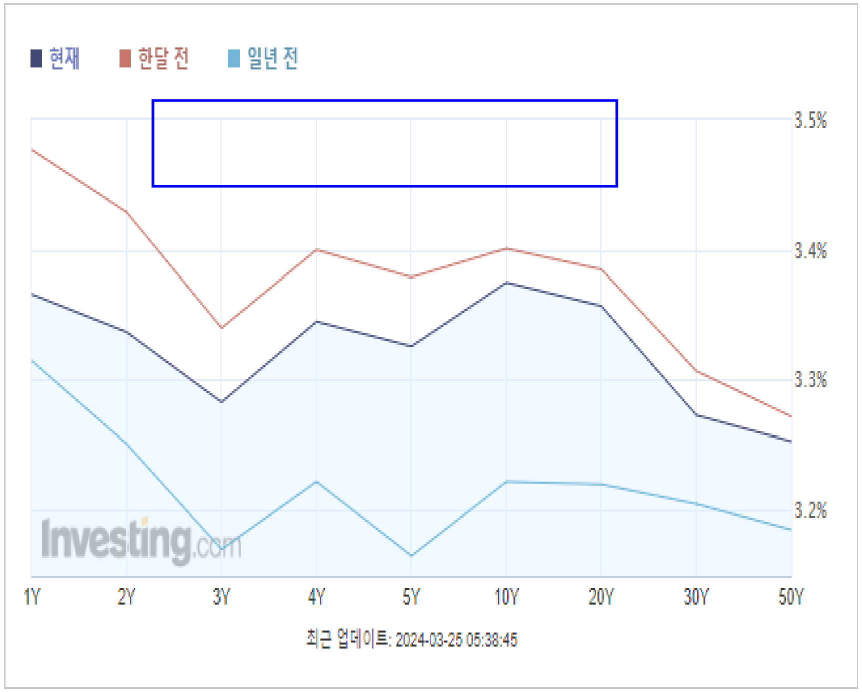
미국 채권

수익률 곡선



한국 채권

수익률 곡선



Yield to Maturity and Prices to maturity on Zero-Coupon Bonds

Maturity (years)	Yield to Maturity (%)	Price
1	5%	$952.38 = $1,000/1.05
2	6	$890.00 = $1,000/1.06^2
3	7	$816.30 = $1,000/1.07^3
4	8	$735.03 = $1,000/1.08^4

TABLE 15.1

Yields and prices to maturity on zero-coupon bonds ($1,000 face value)

Valuing Coupon Bonds

= 3, 10% = 10%, 1000

- Value a 3 year, 10% coupon bond (face value 1000) using Table 15.1:

$$\text{Price} = \frac{\$100}{1.05} + \frac{\$100}{1.06^2} + \frac{\$1100}{1.07^3}$$

-> 3 = 6.88

- Price = $1082.17 and YTM = 6.88%
- 6.88% is less than the 3-year rate of 7%

Yield Curve Under Certainty

The Yield Curve and Future Interest Rates

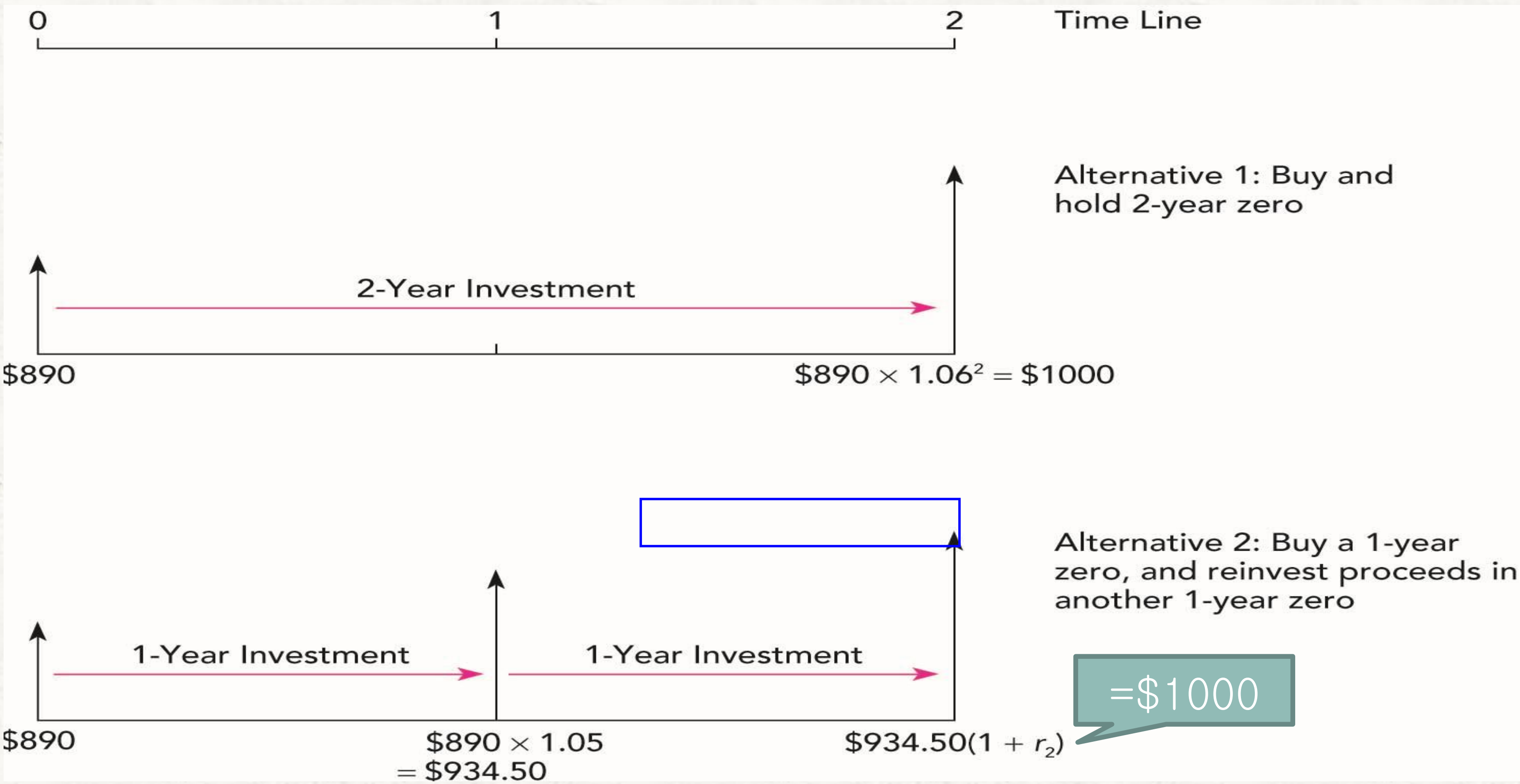
- Yield Curve Under Certainty
  - Suppose you want to invest for 2 years
    - **Buy and hold** a 2-year zero coupon bond

or

    - **Rollover** a series of 1-year zero coupon bonds

  - Equilibrium requires that both strategies provide the same return

2-year investment programs



=$1000

$$(1+0.05)\times(1+r_2)=1.06^2;\ r_2=0.07$$

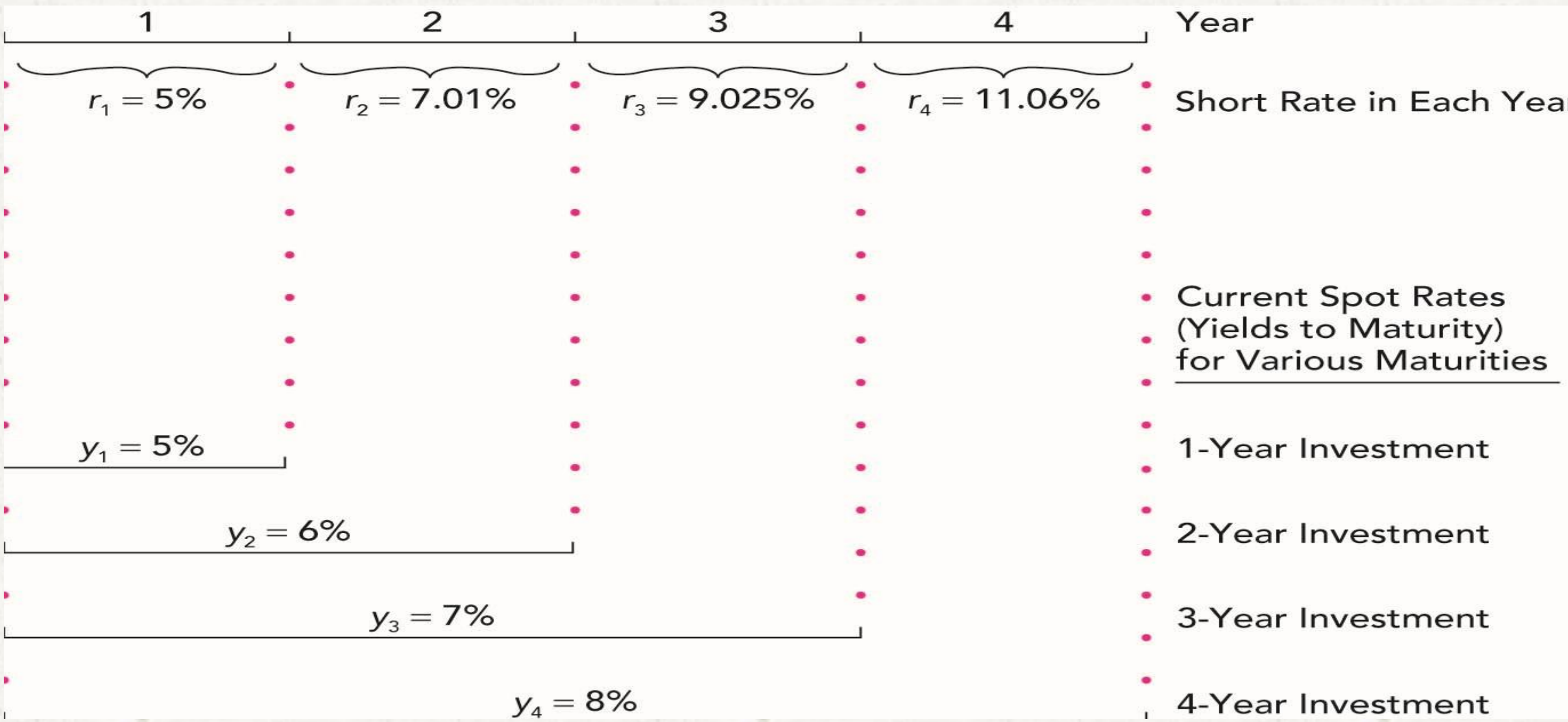
- Yield Curve Under Certainty
  - Buy and hold vs. rollover:

$$\overset{\text{buy and hold}}{(1+y_2)^2} = \overset{\text{rollover}}{(1+r_1)(1+r_2)}$$

$$1+y_2 = [(1+r_1)(1+r_2)]^{\frac{1}{2}}$$

  - Next year's 1-year rate (r_2) is just enough to make rolling over a series of 1-year bonds equal to investing in the 2-year bond

Short Rates vs. YTM



$$(1+y_T)^T = [(1+r_1)(1+r_2)\cdots(1+r_T)]$$

Short Rates and Yield Curve Slope

- When next year's short rate, r_2 , is greater than this year's short rate, r_1, the yield curve **slopes up**
  - May indicate **short rates are expected to rise**

- When next year's short rate, r_2 , is less than this year's short rate, r_1, the yield curve **slopes down**
  - May indicate **short rates are expected to fall**

선도이자율(Forward Rates)

foward: ;

- 선도이자율: **미래 단기이자율(future short rate)의 시장 예상치**로서, 미래 시점에 발행될 유가증권의 **내재수익률(implied rate of return)**을 의미

- 선도이자율 $_if_j$ 에서 i는 현재부터 미래 발행시점까지의 기간을, j는 만기를 의미

$_1f_2$: 1년후 발행될 2년만기 채권의 내재수익률

$_3f_1$: 3년후 발행될 1년만기 채권의 내재수익률

Yield curve and forward rates

1) 수익률곡선이 우상향(upward-sloping)하는 경우, 미래 단기이자율이 현재 단기이자율보다 높아질 것으로 기대

예: y_1=8%, y_2=8.995%인 경우

(1+y2)^2 = (1+r1)(1+1f1)

$$ {}_1f_1 = \frac{(1+y_2)^2}{(1+y_1)} - 1 = 0.10;10\% $$

2) 수익률곡선이 수평(flat)인 경우, 미래 단기이자율이 현 수준에서 불변일 것으로 기대

예: y_1=y_2=8%인 경우

$$ {}_1f_1 = \frac{(1+y_2)^2}{(1+y_1)} - 1 = 0.08;8\% $$

3) 수익률곡선이 우하향(downward-sloping)하는 경우, 미래 단기이자율이 현재보다 낮아질 것으로 기대

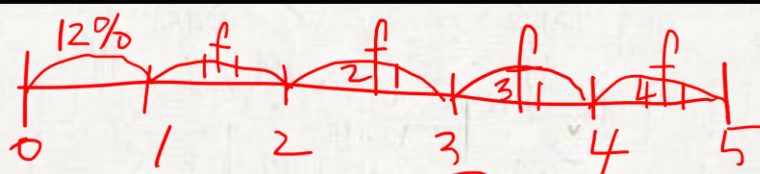
예: y_1=8%, y_2=6%인 경우

$$ {}_1f_1 = \frac{(1+y_2)^2}{(1+y_1)} - 1 = 0.0404;4.04\% $$

Forward Rates with Downward Sloping YC

=> headge-> / ( )

Zero-Coupon Rates	Bond Maturity
12%	1
11.75%	2
11.25%	3
10.00%	4
9.25%	5



만기: 1yr Forward Rates
발행시점

$_1f_1$	1년 후	$[(1.1175)^2 / 1.12] - 1$	=	0.115006	11.5%
$_2f_1$	2년 후	$[(1.1125)^3 / (1.1175)^2] - 1$	=	0.102567	10.26%
$_3f_1$	3년 후	$[(1.1)^4 / (1.1125)^3] - 1$	=	0.063336	6.33%
$_4f_1$	4년 후	$[(1.0925)^5 / (1.1)^4] - 1$	=	0.063008	6.30%

Forward Rates

- 균형조건: 1년 후 발행되는 2년만기 채권을 만기보유(buy&hold)하는 것은 1년 후 발행되는 1년만기 채권에 투자한 후 다시 1년만기 채권에 재투자(rollover)하는 것과 동일한 수익률을 얻어야 함.

(buy&hold) = (rollover)

$$(1+{}_1f_2)^2 = (1+{}_1f_1)(1+{}_2f_1)$$

Why?

$$(1+{}_1f_2)^2 = \frac{(1+y_3)^3}{(1+y_1)} = \frac{(1+y_2)^2(1+y_3)^3}{(1+y_1)(1+y_2)^2} = (1+{}_1f_1)(1+{}_2f_1)$$

Similarly,

$$(1+{}_1f_3)^3 = \frac{(1+y_4)^4}{(1+y_1)} = \frac{(1+y_2)^2(1+y_3)^3(1+y_4)^4}{(1+y_1)(1+y_2)^2(1+y_3)^3} = (1+{}_1f_1)(1+{}_2f_1)(1+{}_3f_1)$$

Generally,

$$1+{}_if_j = \left[\frac{(1+y_{i+j})^{i+j}}{(1+y_i)^i}\right]^{\frac{1}{j}}$$

Example

- 다음과 같은 할인채(액면금액 1000) 가격에 기초하여 선도이자율을 계산하라.

잔존만기	1	2	3	4
가격	943.4	898.47	847.62	792.16

1) 이자율 기간구조:

채권 1: 943.4 = $\frac{1000}{1+y_1}$ => y_1 = 0.060

채권 2: 898.47 = $\frac{1000}{(1+y_2)^2}$ => y_2 = 0.055

채권 3: 847.62 = $\frac{1000}{(1+y_3)^3}$ => y_3 = 0.057

채권 4: 792.16 = $\frac{1000}{(1+y_4)^4}$ => y_4 = 0.060

$${}_1f_1 = \frac{(1+y_2)^2}{1+y_1} - 1 = 0.05$$

$${}_2f_1 = \frac{(1+y_3)^3}{(1+y_2)^2} - 1 = 0.06$$

$${}_3f_1 = \frac{(1+y_4)^4}{(1+y_3)^3} - 1 = 0.07$$

Example

- 할인채 가격에서 계산되는 선도이자율이 다음과 같은 패턴을 보이는 경우,

  ${}_0f_1 = y_1 = 5\%$, ${}_1f_1 = 7\%$, ${}_2f_1 = 8\%$

1) 이자지급이 연 60인 액면 1000, 3년만기 채권의 현재 가격(P_0)과 만기 수익률(YTM)을 구하라.

P_0 = 60/1.05 + 60/[(1.05)(1.07)]
+1060/[(1.05)(1.07)(1.08)]
= 984.14

$$984.14 = \sum_{t=1}^{3} \frac{60}{(1+y_3)^t} + \frac{1000}{(1+y_3)^3}$$

$$y_3 = 6.6\%; YTM$$

y2 =?
(1+0.05)(1+0.07) = (1+y2)^2

2) 실현된 복리수익률 을 구하라.

$$FV = 60 \times 1.07 \times 1.08 + 60 \times 1.08 + 1060 = 1194.14$$

$$y(\text{실현된복리수익률}) = \left(\frac{1194.14}{984.14}\right)^{\frac{1}{3}} - 1 = 0.0666;6.66\%$$

3) 단기(1년) 이자율이 7%로 고정될 것으로 예상하는 경우, 이표채의 1년 보유 기대수익률을 구하라.

5% -> 7% -> 7%

$$P_1 = \frac{60}{1.07} + \frac{1060}{(1.07)^2} = 981.92$$

$$E(\text{보유수익률}) = \frac{[60 + (981.92 - 984.14)]}{984.14} = 0.0587;5.87\%$$

Lock-in loan rates

1년후 차입 ; 만기 1년

- 현 시점에서 미래 차입이자율(loan rates)을 확정(lock-in)시키는 2가지 거래유형

이자율위험 헤지수단!!

거래 1) 할인채(액면금액 1000)의 시장가격들을 이용하여, 1년 후 차입할 1년 만기 대출이자율을 확정:

만기	1	2
채권가격	952.38	890

y_1

$$890 = \frac{1000}{(1+y_2)^2}$$

- 만기수익률을 계산,
  $y_1=5\%$, $y_2=6\%$

$$952.38 = \frac{1000}{(1+y_1)} \Rightarrow y_1 = 0.05$$

- 선도이자율(미래 단기이자율 예상치)을 계산,

$${}_1f_1 = \frac{(1+y_2)^2}{1+y_1} - 1 = 0.0701; 7.01\%$$

forward rate ?

거래 2) 채권거래(1년만기 할인채 매입 + 2년만기 채권 공매도)를 통해 1년 후 차입할 1년만기 대출이자율을 확정:

t =	0 현재	1 차입	2 상환
Buy (1-year bond)	-952.38	1000	
공매도 Short Sell (2-year bond)	952.38 = 890x1.0701		-1000x1.0701
Cash flow	0	1000	-1070.10

1년후 1년만기 차입시 상환원리금: $1000 \times (1+r_2) = 1070.10$

미래 대출이자율: $r_2 = 7.01\%$로 확정 (선도이자율수준과 동일)

- 주의: 현 시점에서 선도이자율로 고정시키는 것이 미래 단기이자율이 현재 선도이자율과 같아진다는 의미는 아님.

$_1f_1 = 0.0701$

8% 유리

6% 불리

선도계약 (forward contracts)

-> ()

- 선도계약: 현재 약정한 인도가격(delivery price)으로 특정 미래시점에 유가증권을 매매한다는 약속

- 일반적으로 선도계약은 거래소를 통하지 않으며(장외시장: OTC market), 외환시장, 채권시장 등에서 널리 이용됨.

예) 이자율 기간구조가 다음과 같고, 2년 후 2년만기 **할인채**(액면 1000)를 매입하는 선도계약의 경우,

만기수익률:

$y_1 = 6\%,\ y_2 = 6.5\%,\ y_3 = 7\%,\ y_4 = 7.5\%$

$$\text{선도가격}: P(forward) = \frac{1000}{(1+{}_2f_2)^2}$$

$$\text{선도이자율}: {}_2f_2 = \left[\frac{(1+y_4)^4}{(1+y_2)^2}\right]^{\frac{1}{2}} - 1 = 0.0851$$

$$\Rightarrow P(forward) = 849.3$$

Exercise

- 이자율 기간구조가 다음과 같을 때, 2년 후 발행되는 2년만기 **이표율 10% 채권**(액면 1000)의 선도가격을 구하라.

$y_1 = 6\%,\ y_2 = 6.5\%,\ y_3 = 7\%,\ y_4 = 7.5\%$

$$\text{힌트}: P(forward) = \frac{100}{1+{}_2f_1} + \frac{1100}{(1+{}_2f_2)^2}$$

$$\text{답}: P(forward) = 1026.8$$

Yield Curve Under Uncertainty

Interpreting the Term Structure

- The yield curve reflects **expectations of future interest rates**
- The forecasts of future rates are clouded by other factors, such as liquidity premiums
- An **upward sloping yield curve** indicates:
  - Short term rates are **expected to rise**

  and/or

  - Investors require **liquidity premiums** to hold long term bonds
- The yield curve is a good predictor of the business cycle
  - Long term rates tend to **rise** in anticipation of economic **expansion**
  - Inverted yield curve may indicate that interest rates are expected to fall and signal a recession

Interest Rate Uncertainty

- Suppose that today's short term rate is 5% and the *expected* short rate for the following year is $E(r_2)$ = 6%. The expected value of a 2-year zero coupon bond (face value, 1000) is:

$$\frac{\$1000}{(1.05)(1.06)} = \$898.47$$

- The value of a 1-year zero coupon bond (face value, 1000):

$$\frac{\$1000}{1.05} = \$952.38$$

Interest Rate Uncertainty

- The investor wants to invest for 1 year:

Option 1)

Buy the 2-year bond today and sell it at the end of the first year for $1000/1.06 =$943.40 ⇒ 예상수익률 5% ; uncertain !!

$= \frac{943.4 - 898.47}{898.47}$

E(r_2) ; uncertain !!

Option 2)

Buy the 1-year bond today and hold to maturity

⇒ 수익률 5% ; certain !!

$= \frac{1000 - 952.38}{952.38}$

- What if next year's interest rate is more (or less) than 6%? The actual return on the 2-year bond is **uncertain**!

Interest Rate Uncertainty

- Investors require a **risk premium** to hold a longer-term bond.
- This premium compensates short-term investors for the uncertainty about future prices.

이자율 기간구조에 관한 이론

- 기대 가설 이론(Expectations Hypothesis)
- 유동성 선호 이론(Liquidity Preference)
- 시장 분할 이론(Market Segmentation)
- 선호영역 이론(Preferred Habitat)

1. 기대 가설 이론 (Expectations Hypothesis)

- 가정

– 위험 중립적 투자자

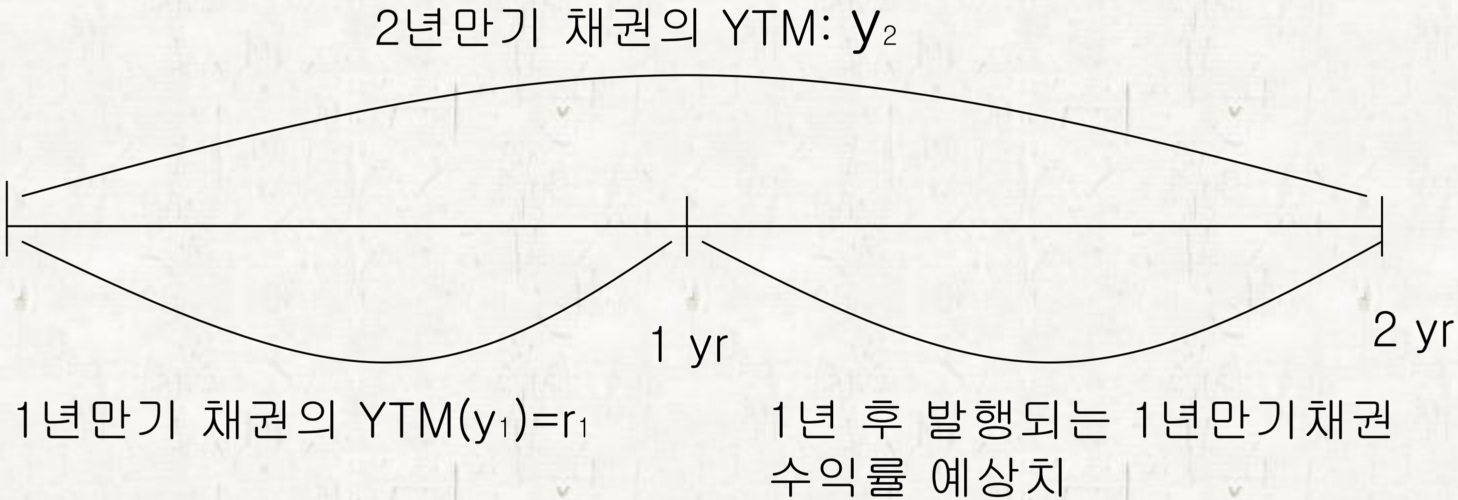
– 장기 채권과 단기 채권은 완전 대체재

– 장기채권의 수익률은 단기채권을 상환연장(roll-over)하여 재투자하는 방식의 예상 수익률(기하평균수익률)과 동일

- 현재 채권의 만기수익률이 미래 이자율에 대한 투자자들의 예상을 그대로 반영한다는 이론
- The Expectations Hypothesis Theory
  - Observed long-term rate is a function of today's short-term rate and expected future short-term rates
  - ${}_{n-1}f_1 = E(r_n)$ and **liquidity premiums are zero**

기대 가설 이론

- 장기 채권과 단기 채권의 이자율 구조



선도이자율: ${}_1f_1=E(r_2)$

hold

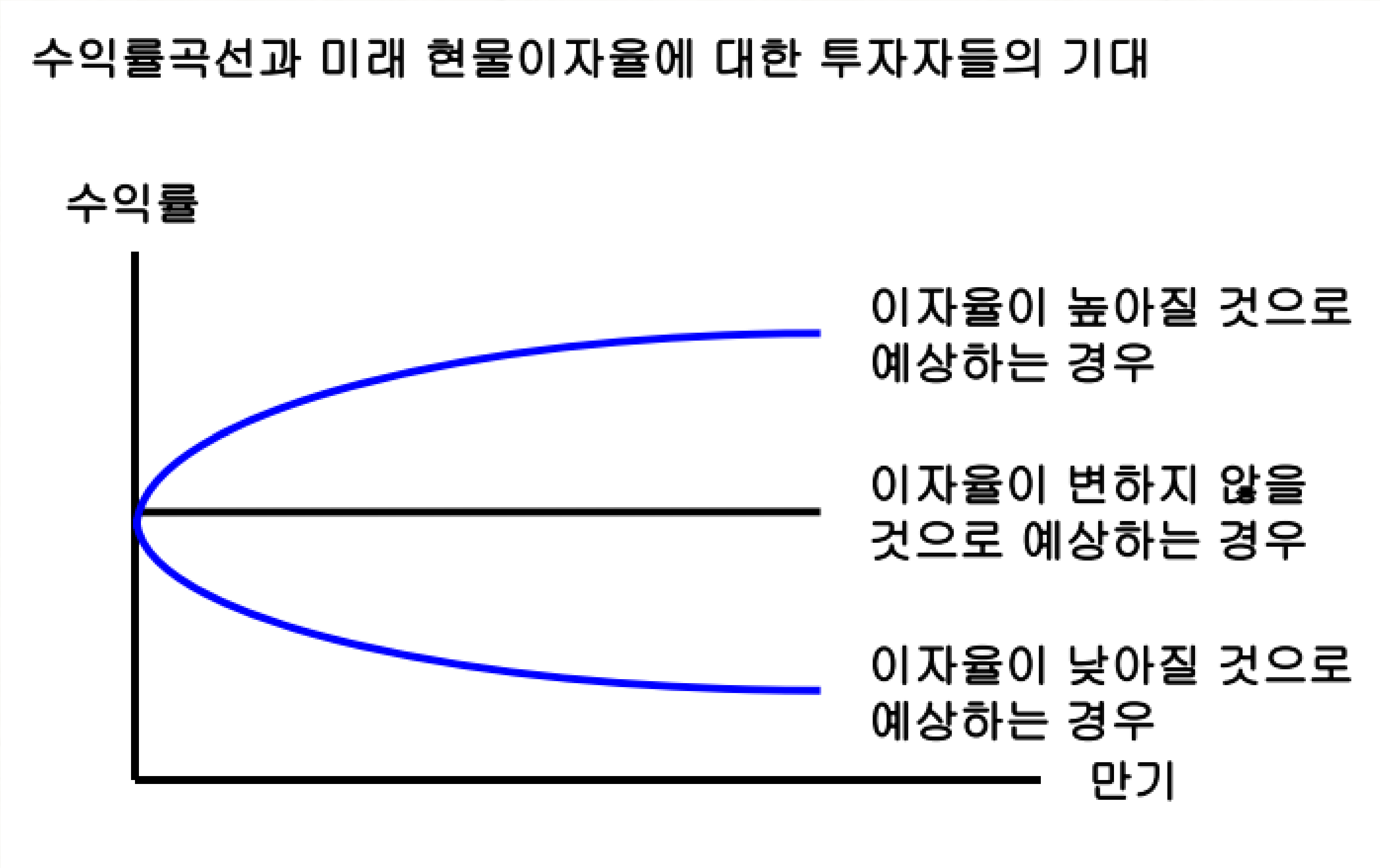
roll up

$$(1+y_2)^2=(1+r_1)(1+E(r_2))$$

n unbiased?

$$\therefore (1+y_n)^n=(1+r_1)(1+E(r_2))\cdots(1+E(r_n))$$

기대 가설 이론


수익률곡선과 미래 현물이자율에 대한 투자자들의 기대
수익률
이자율이 높아질 것으로 예상하는 경우
이자율이 변하지 않을 것으로 예상하는 경우
이자율이 낮아질 것으로 예상하는 경우
만기

2. 유동성 선호 이론 (Liquidity Preference Theory)

- 불확실한 시간에 대한 위험보상으로 추가수익률을 요구
  - 만기가 길수록 유동성프리미엄(liquidity premium)은 증가
  - 시장 주류는 단기투자자

- Forward rate
  = expected future short term interest rate($E(r_i)$)+ **Liquidity premium(l_i)**

∴수정기대가설(modified expectations hypothesis)에 해당

- 가정

- **위험 회피적인 투자자**
- 장기 채권과 단기 채권은 대체재
  - 투자자들이 **단기투자(유동성)을 선호**하므로 장기투자에 대해 프리미엄을 요구

$$\therefore (1+y_n)^n = (1+r_1+l_1)(1+E(r_2)+l_2)\cdots(1+E(r_n)+l_n)$$

유동성 선호 이론

- Long-term bonds are more risky;
  therefore, ${}_{n-1}f_1$ exceeds $E(r_n)$

- The excess of ${}_{n-1}f_1$ over $E(r_n)$ is the *liquidity premium*

- The yield curve has an **upward bias** built into the long-term rates because of the liquidity premium

Upward sloping yield curve with liquidity premium

- Suppose r_1=5% and expected short interest rate $E(r_i)$ is 5% as constant.

$(1+y_2)^2=(1+r_1)(1+E(r_2))=1.05\text{x}1.05$

$y_2=5\%$

If liquidity premium is 1%, ${}_1f_1$=6%.

$(1+y_2)^2=(1+r_1)(1+{}_1f_1)=1.05\text{x}1.06=1.113$

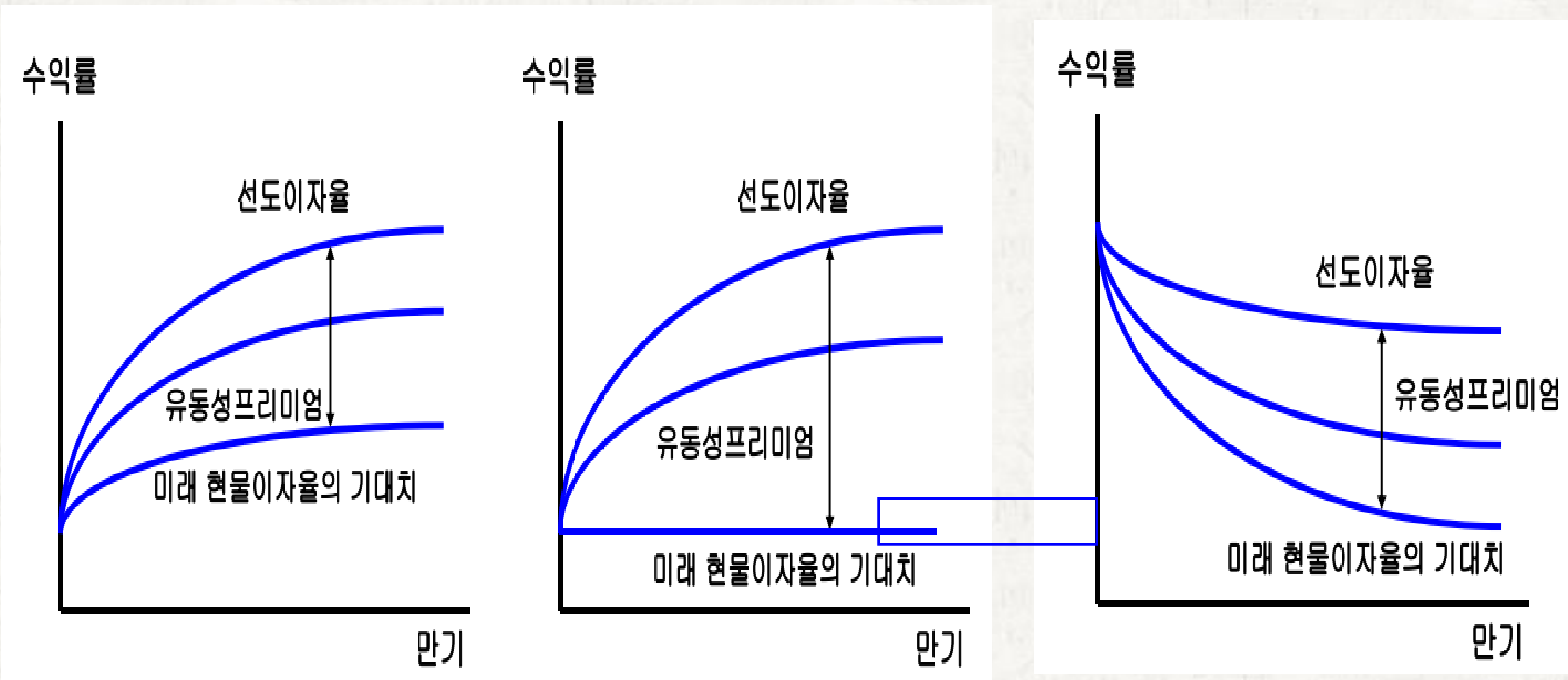
$y_2=5.5\%$

$(1+y_3)^3=(1+r_1)(1+{}_1f_1)(1+{}_2f_1)=1.05\text{x}1.06\text{x}1.06=1.1798$

$y_3=5.67\%$

as $t \rightarrow \infty$, $y_t \approx 6\%$

유동성 선호 이론


수익률
선도이자율
유동성프리미엄
미래 현물이자율의 기대치
만기
수익률
선도이자율
유동성프리미엄
미래 현물이자율의 기대치
만기
수익률
선도이자율
유동성프리미엄
미래 현물이자율의 기대치
만기

3. 시장분할 이론 (Market Segmentation Theory)

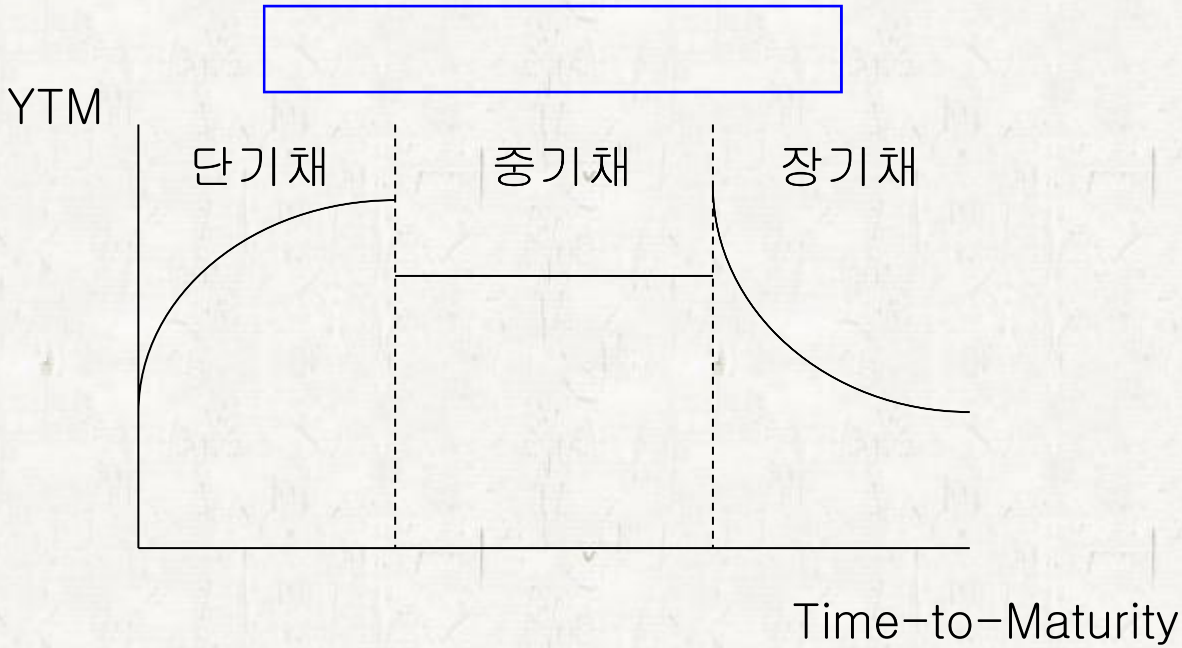
- 채권 시장의 이자율은 만기별로 서로 다른 수요와 공급에 의해서 결정
- 장기 채권 시장의 경우 수요가 적으므로 가격이 낮음(이자율이 높음)
- 수익률곡선(yield curve)은 불연속적

참고) Culberstone, J.M., “The Term Structure of Interest Rates,” QJE, 1957

- 가정

- 시장 참여자들의 채권에 대한 선호가 다르다.
- 장기채권과 단기 채권은 대체재가 아니다.

시장 분할 이론



- 문제점:

  - 만기가 다른 채권의 이자율이 서로 같이 변동하는 것을 설명할 수 없다.

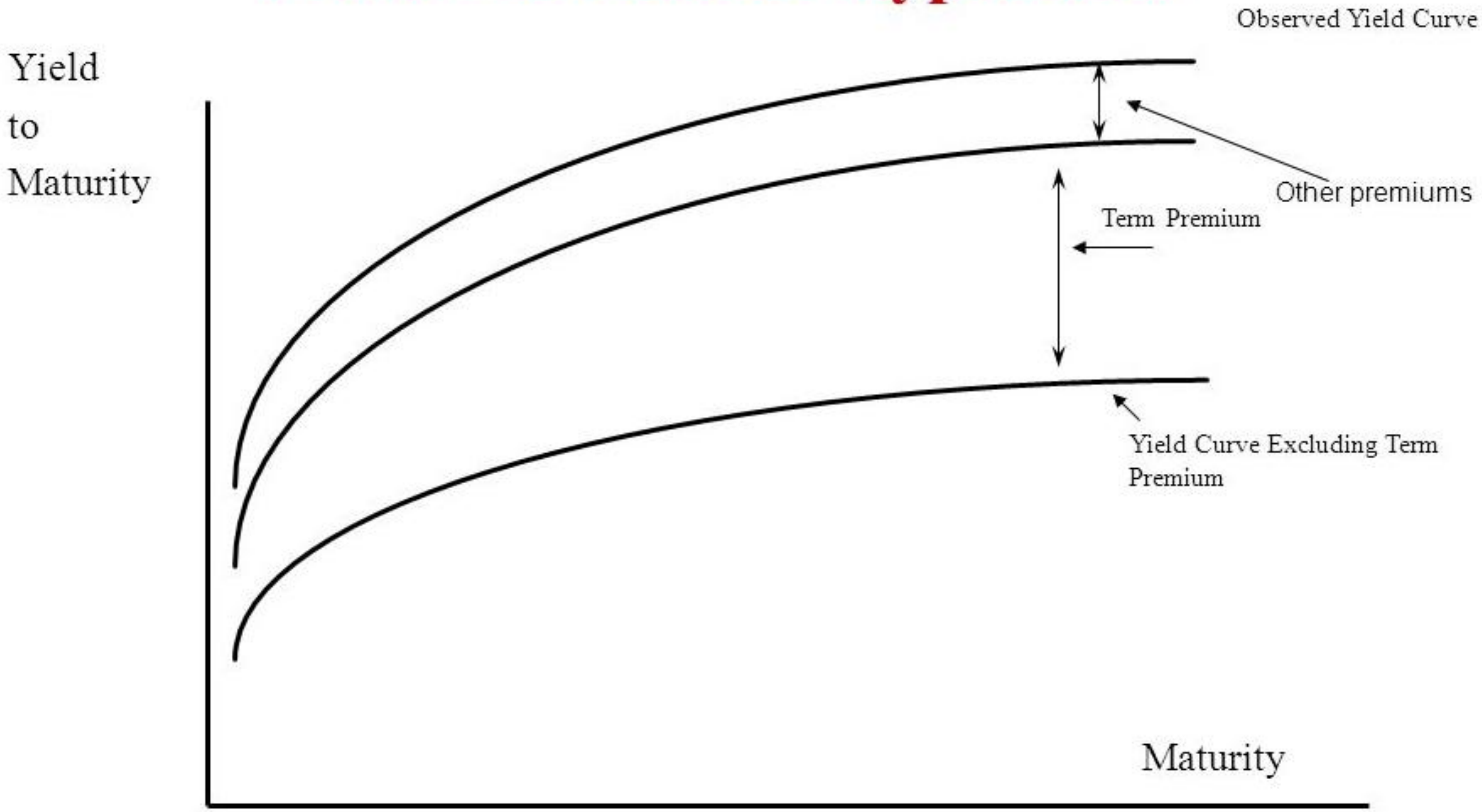
  - **재정거래(arbitrage)로 인해 만기별로 독립적인 상이한 수익률곡선은 형성될 수 없다.**

4. 선호영역 이론 (Preferred Habitat Theory)

- 투자자별로 선호하는 특정 만기(영역)가 존재하지만, 다른 만기채권에 대한 충분한 **보상(habitat premium)**이 주어지면 투자할 수 있다.
- 기대이론, 유동성 프리미엄 이론, 시장분할 이론의 3가지 이론을 **절충**한 이론
- 프리미엄이 충분하지 않을 경우 선호하는 만기채권에만 투자하기 때문에 장단기채권시장은 부분적으로 분할
- 따라서 다양한 만기를 가진 채권수익률에 있어서의 모순은 장단기채권 시장에서의 재정거래에 의해서 완전히 제거되지는 못함.

참고) Modigliani and Sutch, “Debt Management and the Term Structure of Interest Rates: An Empirical Analysis of Recent Experience, JPE, 1967

Graphic Portrayal of the Preferred Habitat Hypothesis



채권수익률의 위험구조

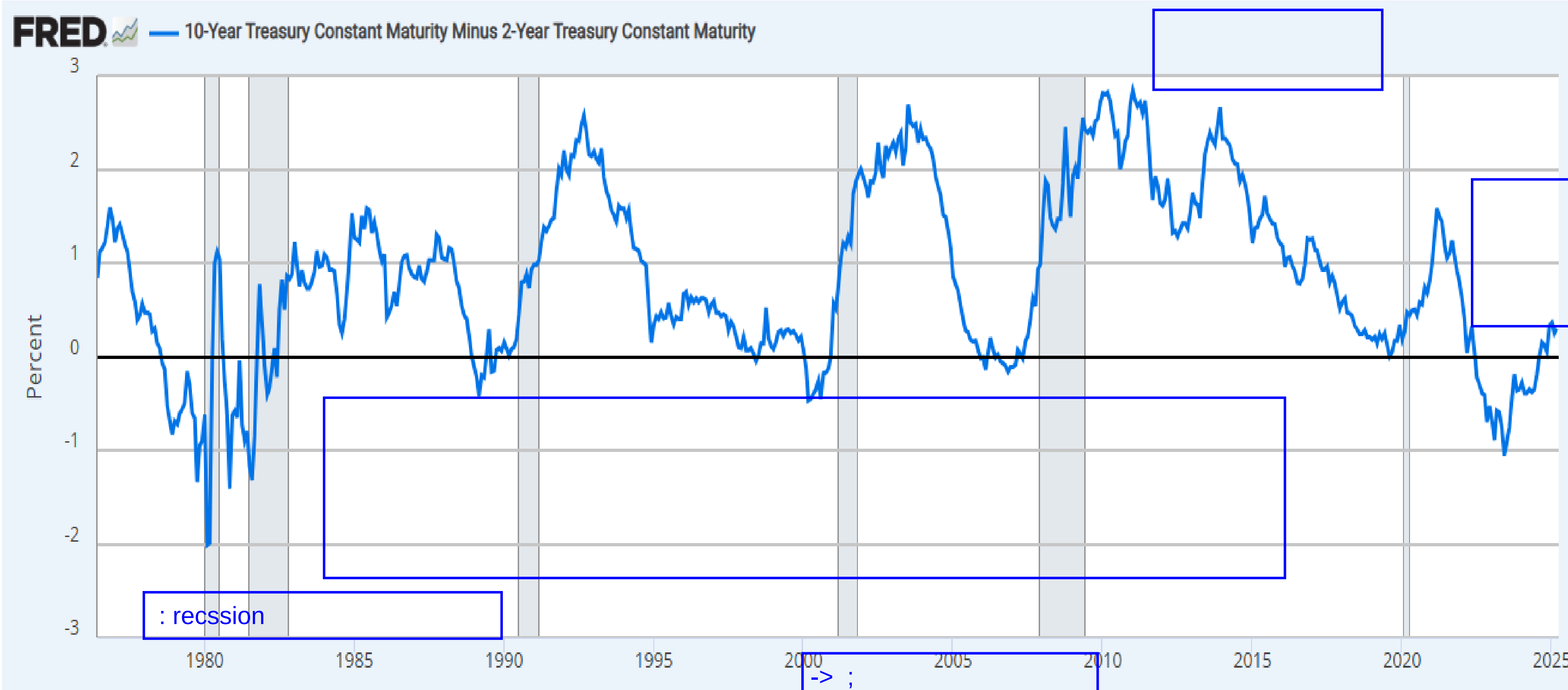
A yield spread is the difference between yields on differing debt instruments of varying **maturities, issuers, or risk levels**.

- 채권의 발행조건이나 발행자의 신용도 차이로 인해 채권의 수익률에 체계적인 차이가 나타나는 것을 채권수익률의 위험구조(risk structure of yields)라고 함.

Term spread

10-Year Treasury Constant Maturity Minus 2-Year Treasury Constant Maturity (T10Y2Y)

Observations	Units:	Frequency:		
2025-03-24: **0.30**	Percent,	Daily	1Y 5Y 10Y Max	Edit Graph
Updated: Mar 24, 2025 4:02 PM CDT	Not Seasonally Adjusted		1976-06-01 to 2025-03-24	Download
Next Release Date: Mar 25, 2025				



Source: Federal Reserve Bank of St. Louis via FRED®

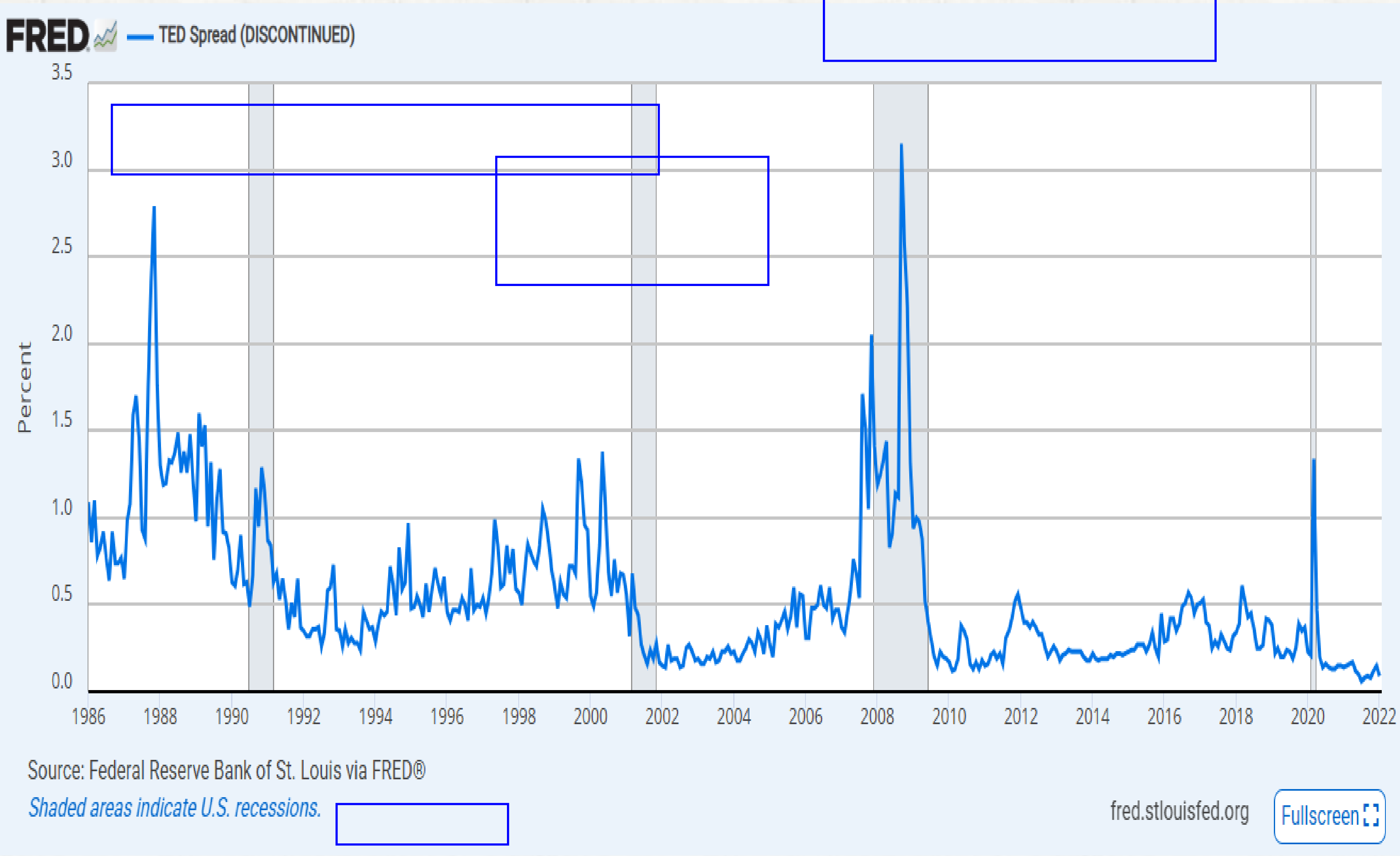
Shaded areas indicate U.S. recessions.

fred.stlouisfed.org

Fullscreen

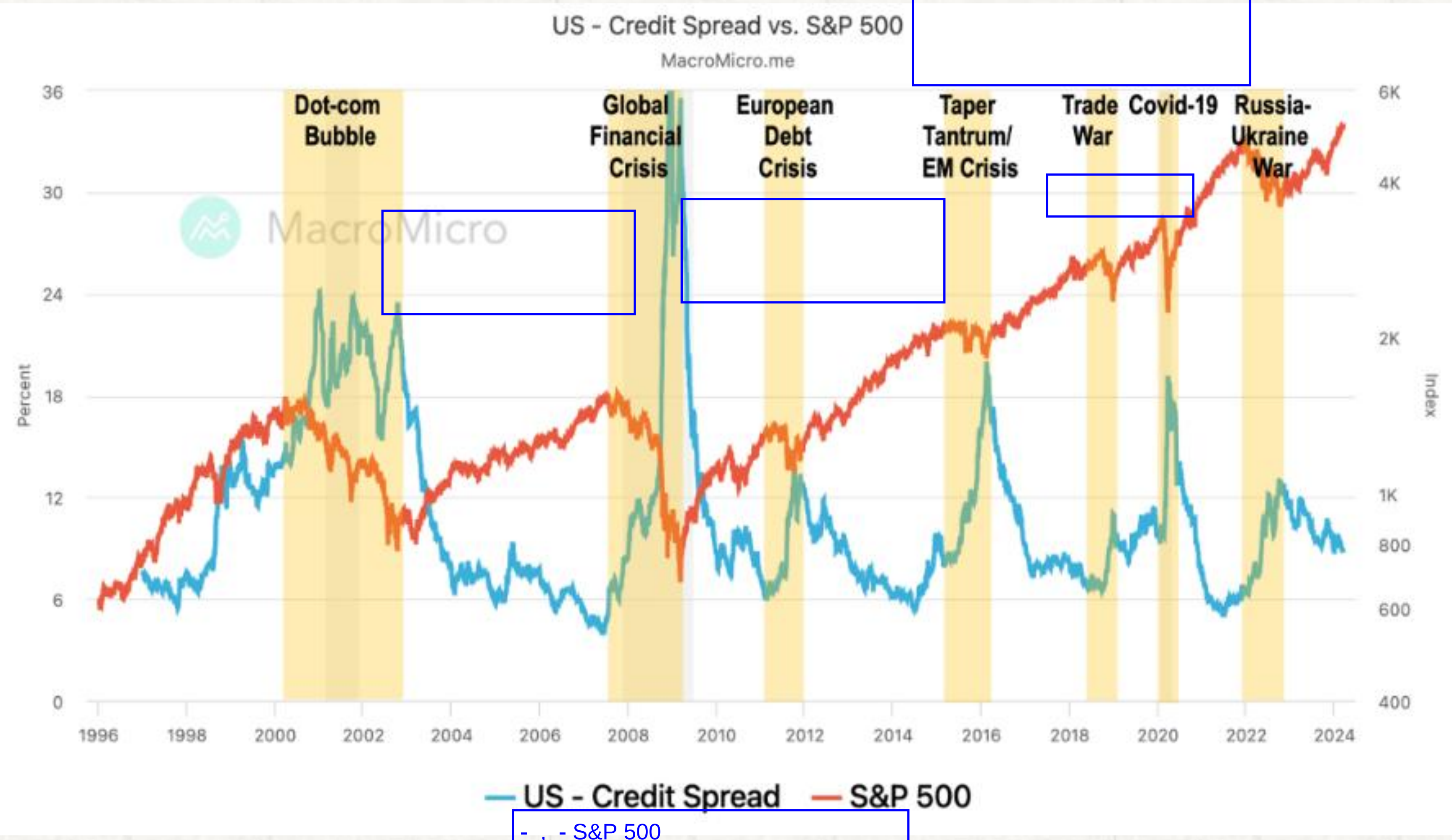
TED Spread

between 3-Month LIBOR based on US dollars and 3-Month Treasury Bill (The discontinuation of LIBOR in 2021 led to its replacement by the Secured Overnight Financing Rate (SOFR))



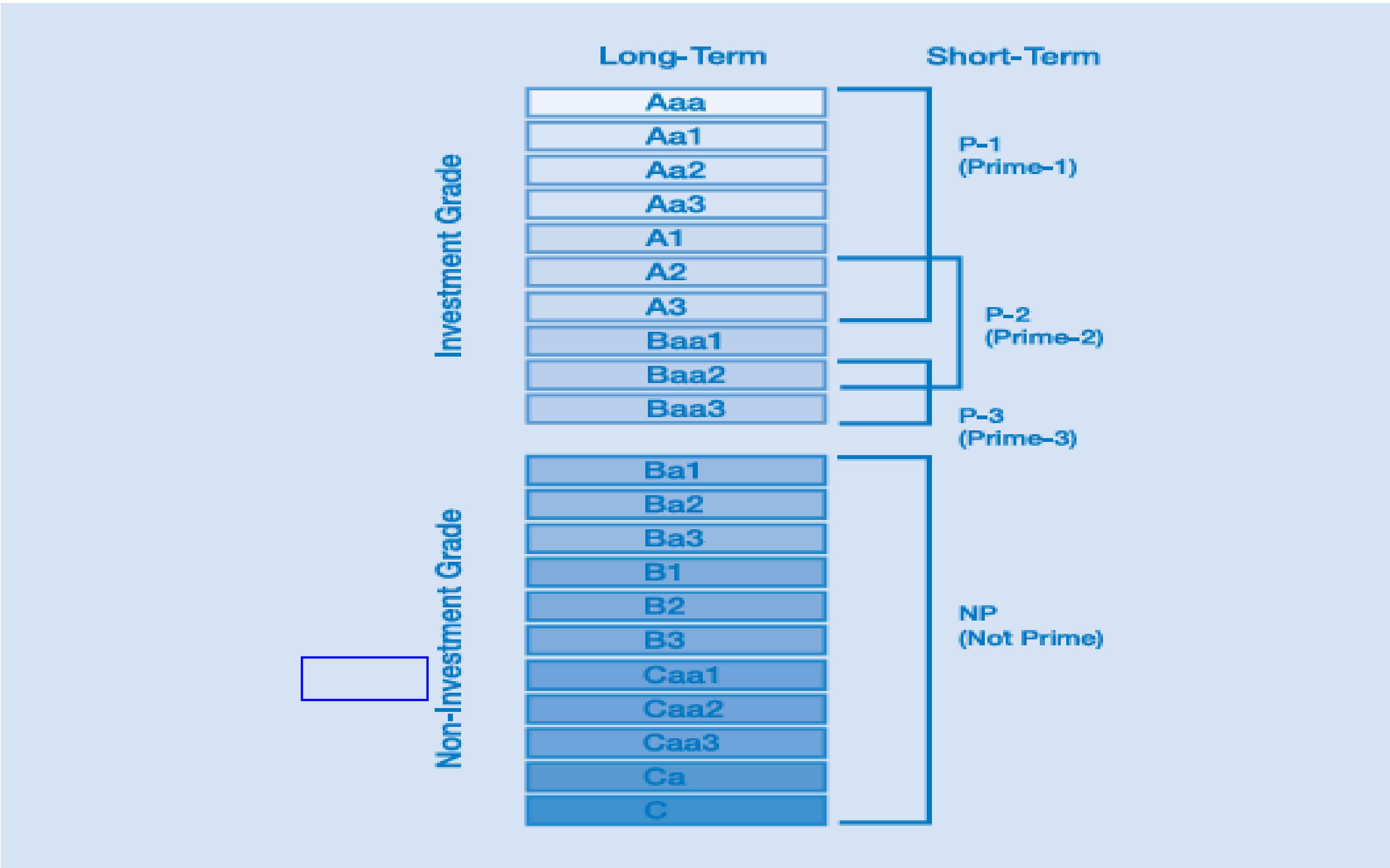
Credit spread

The credit spread shown in the chart is calculated as yield on Bofa Merrill Lynch CCC-rated high-yield bonds minus the 10-year Treasury yield.



Moody's Rating Scale

The following is a ranking (from highest to lowest) of Moody's long-term and short-term categories. The indicated relationship between long-term and short-term ratings is approximate and may not necessarily apply in all situations.



Moody's Short-Term Rating Definitions

Moody's short-term ratings, unlike our long-term ratings, apply to an individual issuer's capacity to repay all short-term obligations rather than to specific short-term borrowing programs. Once assigned to an issuer, a short-term rating is global in scope; it applies to all the issuer's senior, unsecured obligations with an original maturity of less than one year regardless of the currency or market in which the obligations are issued. An exception to the global nature of these ratings occurs if an issuer's rating is supported by another entity through vehicles such as a letter of credit or guarantee.

Moody's employs the following designations, all judged to be investment grade, to indicate the relative repayment ability of rated issuers:

P-1	Issuers (or supporting institutions) rated Prime-1 have a superior ability to repay short-term debt obligations.
P-2	Issuers (or supporting institutions) rated Prime-2 have a strong ability to repay short-term debt obligations.
P-3	Issuers (or supporting institutions) rated Prime-3 have an acceptable ability to repay short-term obligations.
NP	Issuers (or supporting institutions) rated Not Prime do not fall within any of the Prime rating categories.

Moody's Long-Term Rating Definitions

Moody's long-term obligation ratings are opinions of the relative credit risk of fixed-income obligations with an original maturity of one year or more. They address the possibility that a financial obligation will not be honored as promised. Such ratings reflect both the likelihood of default and any financial loss suffered in the event of default.

Aaa	Obligations rated Aaa are judged to be of the highest quality, with minimal risk.
Aa	Obligations rated Aa are judged to be of high quality and are subject to very low credit risk.
A	Obligations rated A are considered upper-medium-grade and are subject to low credit risk.
Baa	Obligations rated Baa are subject to moderate credit risk. They are considered medium-grade and as such may possess speculative characteristics.
Ba	Obligations rated Ba are judged to have speculative elements and are subject to substantial credit risk.
B	Obligations rated B are considered speculative and are subject to high credit risk.
Caa	Obligations rated Caa are judged to be of poor standing and are subject to very high credit risk.
Ca	Obligations rated Ca are highly speculative and are likely in, or very near, default, with some prospect of recovery in principal and interest.
C	Obligations rated C are the lowest-rated class of bonds and are typically in default, with little prospect for recovery of principal and interest.

S&P credit rating

INVESTMENT GRADE

AAA — **AA** — **A** → **BBB**

AAA
Extremely strong capacity to meet financial commitments

AA
Very strong capacity to meet financial commitments

A
Strong capacity to meet financial commitments, but somewhat susceptible to economic conditions and changes in circumstances

BBB
Adequate capacity to meet financial commitments, but more subject to adverse economic conditions

SPECULATIVE GRADE

BB — **B** — **CCC** — **CC** — **C** → **D**

BB
Less vulnerable in the near-term but faces major ongoing uncertainties to adverse business, financial and economic conditions

B
More vulnerable to adverse business, financial and economic conditions but currently has the capacity to meet financial commitments

CCC
Currently vulnerable and dependent on favorable business, financial and economic conditions to meet financial commitments

CC
Highly vulnerable; default has not yet occurred, but is expected to be a virtual certainty

C
Currently highly vulnerable to non-payment, and ultimate recovery is expected to be lower than that of higher rated obligations

D
Payment default on a financial commitment or breach of an imputed promise; also used when a bankruptcy petition has been filed